

BBPS

Bharat Bill Payment System

Partner API Integration Guide

Eko India Financial Services

API Version: v3

Base URL: <https://api.eko.in:25002/ekoicici/v3>

Classification: Confidential — for integrating partners

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1. Overview

This guide describes how to integrate the **Bharat Bill Payment System (BBPS)** service exposed by Eko's `ekoicici/v3` API. BBPS lets your agents pay utility and recurring bills — electricity, water, gas, broadband, DTH, mobile, FASTag, loans, insurance, and more — and perform mobile/DTH recharges on behalf of walk-in customers.

The guide covers the full flow: discovering billers (operators) by category and location, reading the parameters an operator requires, fetching a live bill amount, and paying the bill. Recharge-specific lookups (operator detection and plan listing) are also documented. Each endpoint section lists the request parameters, a sample request, and the success and error responses your integration should handle.

Base URL: **`https://api.eko.in:25002/ekoicici/v3`**

Intended audience

Partner engineering teams integrating BBPS into their own applications. Familiarity with REST APIs and form-encoded / JSON requests is assumed.

Key concepts

- **Operator (biller)** — the entity being paid, identified by an `operator_id`. Passed as `phone_operator_code` at bill-fetch / pay time.
- **Category** — the biller type (Electricity, Water, DTH, etc.), identified by `operator_category_id`.
- **Location** — the operating region (state), identified by `operator_location_id`.
- **Operator parameters** — the input fields a given operator requires (e.g. account number, mobile number), returned per operator.
- **client_ref_id** — a unique partner reference sent on bill-fetch and pay for idempotency and reconciliation.

Typical bill-payment sequence

- **Get Categories / Get Locations** — optional lookups to filter billers.
- **Get Operators** — list billers (optionally by category and/or location).
- **Get Operator Parameters** — read the fields the chosen operator requires.
- **Fetch District Discome** — only for operator 190 (UPPCL), to resolve `district_discome`.
- **Fetch Bill** — retrieve the live bill amount and customer name.
- **Pay Bill** — pay the fetched amount; store the returned `tid`.

Recharge sequence (prepaid mobile / DTH)

- **Get Recharge Operator** — detect operator and circle from the customer mobile number.
- **Get Recharge Plans** — list available plans for the operator and circle.
- **Pay Bill** — submit the recharge as a payment.

Common request headers

Header	Description
<code>developer_key</code>	Your developer key, issued by Eko during onboarding.
<code>secret-key</code>	Request signature, computed per the Eko signing specification.
<code>secret-key-timestamp</code>	Epoch timestamp (ms) used to generate the secret-key.
<code>Content-Type</code>	<code>application/x-www-form-urlencoded</code> or <code>application/json</code> .

Request format

All endpoints accept either `application/x-www-form-urlencoded` or `application/json`, with identical field names. Lookup (GET) endpoints take their parameters in the query string.

2. Discovery & Lookups

These read-only endpoints let you build biller selection screens: list categories, list locations, and list operators (optionally filtered). Use the returned identifiers to drive the parameter and bill-fetch calls.

2.1 Get Categories

GET /customer/payment/bbps/categories?initiator_id={initiator_id}

Returns all BBPS biller categories.

Sample request

```
curl --location 'https://api.eko.in:25002/ekoicici/v3/customer/payment/bbps/categories?initiator_id=<initiator_id>'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
--header 'secret-key-timestamp: <timestamp>'
```

Success response (list truncated)

```
{
  "response_status_id": 0,
  "param_attributes": {
    "list_elements": [
      { "operator_category_name": "Broadband Postpaid", "operator_category_id": 1,
"operator_category_group": "0", "status": "1" },
      { "operator_category_name": "Electricity", "operator_category_id": 8,
"operator_category_group": "0", "status": "1" },
      { "operator_category_name": "Water", "operator_category_id": 11,
"operator_category_group": "0", "status": "1" },
      { "operator_category_name": "FASTag", "operator_category_id": 22,
"operator_category_group": "0", "status": "1" }
      /* ... additional categories ... */
    ]
  },
  "response_type_id": 2457,
  "message": "BBPS category fetch success",
  "status": 0
}
```

Response fields

Field	Description
operator_category_id	Integer category identifier — pass as category to Get Operators.
operator_category_name	Human-readable category name.
status	1 = active.

2.2 Get Locations

GET /customer/payment/bbps/locations?initiator_id={initiator_id}

Returns all operating locations (states / UTs).

Sample request

```
curl --location 'https://api.eko.in:25002/ekoicici/v3/customer/payment/bbps/locations?initiator_id=<initiator_id>'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
```

```
--header 'secret-key-timestamp: <timestamp>'
```

Success response (list truncated)

```
{
  "response_status_id": 0,
  "param_attributes": {
    "list_elements": [
      { "operator_location_name": "Haryana",          "operator_location_id": "06", "abbreviation":
"HR" },
      { "operator_location_name": "Madhya Pradesh",  "operator_location_id": "23", "abbreviation":
"MP" },
      { "operator_location_name": "Andaman and Nicobar", "operator_location_id": "35",
"abbreviation": "AN" }
      /* ... additional locations ... */
    ]
  },
  "response_type_id": 2459,
  "message": "BBPS location fetch success",
  "status": 0
}
```

Response fields

Field	Description
operator_location_id	Location identifier — pass as location to Get Operators.
operator_location_name	State / UT name.
abbreviation	Two-letter code.

2.3 Get Operators

GET /customer/payment/bbps/operators?initiator_id={initiator_id}

Returns the list of billers. Optionally filter by category, location, or both, using the identifiers from Sections 2.1 and 2.2.

Query parameters

Parameter	Type	Required	Description
initiator_id	string	Yes	Partner initiator ID.
category	integer	No	operator_category_id from Get Categories.
location	integer	No	operator_location_id from Get Locations.

Sample requests

```
# All operators
curl --location 'https://api.eko.in:25002/ekoicici/v3/customer/payment/bbps/operators?
initiator_id=<initiator_id>' --header 'developer_key: <developer_key>' --header 'secret-key:
<secret-key>' --header 'secret-key-timestamp: <timestamp>'

# By category
.../bbps/operators?initiator_id=<initiator_id>&category=11

# By location
.../bbps/operators?initiator_id=<initiator_id>&location=35

# By both category and location
.../bbps/operators?initiator_id=<initiator_id>&category=11&location=35
```

Success response (list truncated)

```

{
  "response_status_id": 0,
  "param_attributes": {
    "list_elements": [
      {
        "operator_id": 159,
        "name": "ACT Fibernet",
        "billFetchResponse": 0,
        "high_commission_channel": 0,
        "kyc_required": 1,
        "operator_category": 1,
        "location_id": 0
      },
      {
        "operator_id": 541,
        "name": "Port Blair Municipal Council - Water",
        "billFetchResponse": 0,
        "high_commission_channel": 0,
        "kyc_required": 0,
        "operator_category": 11,
        "location_id": 35
      }
    ]
  },
  /* ... additional operators ... */
  "response_type_id": 2461,
  "message": "BBPS operators fetch success",
  "status": 0
}

```

Response fields

Field	Description
operator_id	Biller identifier — pass as phone_operator_code to Fetch Bill / Pay Bill.
name	Biller name.
operator_category	Category this operator belongs to.
location_id	Location this operator serves (0 = pan-India).
kyc_required	Whether customer KYC is required for this operator.
billFetchResponse	Whether the operator supports live bill fetch.

Error — invalid / missing initiator_id

```

{
  "response_status_id": 1,
  "invalid_params": { "initiator_id": "Merchant does not exist in system." },
  "data": { "csp_id": "7042769385" },
  "response_type_id": -1,
  "message": "Invalid Sender/Initiator",
  "status": 319
}

```

2.4 Get Operator Parameters

GET /customer/payment/bbps/operator/{operator_id}/parameters?initiator_id={initiator_id}

Returns the input fields the chosen operator requires (with validation regex and labels). Use this to render the bill-entry form dynamically. {operator_id} is the operator_id from Get Operators.

Sample request

```
curl --location 'https://api.eko.in:25002/ekoicici/v3/customer/payment/bbps/operator/{operator_id}/parameters?initiator_id=<initiator_id>'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
--header 'secret-key-timestamp: <timestamp>'
```

Success response

```
{
  "response_status_id": 0,
  "param_attributes": {
    "operator_name": "BSNL Prepaid",
    "list_elements": [
      {
        "param_label": "Mobile Number",
        "param_name": "utility_acc_no",
        "param_id": "1",
        "param_type": "Numeric",
        "regex": "^[0-9]{10}$",
        "error_message": "Please enter a valid 10 digit Mobile Number (eg. 0940763946)"
      },
      {
        "param_label": "Recharge Type",
        "param_name": "Recharge Type",
        "param_id": "99",
        "param_type": "List",
        "regex": "1|3",
        "error_message": "Please enter valid Recharge Type"
      }
    ],
    "operator_id": 5,
    "fetchBill": 0,
    "BBPS": 0
  },
  "response_type_id": 2463,
  "message": "BBPS operator parameters fetch success",
  "status": 0
}
```

Response fields

Field	Description
operator_name	Display name of the operator.
list_elements[].param_name	Field name to submit at bill-fetch / pay time.
list_elements[].param_label	Label to show the agent.
list_elements[].param_type	Field type (Numeric, List, etc.).
list_elements[].regex	Client-side validation pattern.
fetchBill	Whether live bill fetch is supported (1 = yes).

3. District Discome (operator 190)

GET /customer/payment/bbps/operators/190/district-discome?initiator_id={initiator_id}

Some electricity billers require a district-level distribution company (discome) code. This applies to **operator 190 (UPPCL)** only — it is not required for other operators. Resolve the `district_discome` value here before calling Fetch Bill / Pay Bill for operator 190.

Sample request

```
curl --location 'https://api.eko.in:25002/ekoicici/v3/customer/payment/bbps/operators/190/district-discome?initiator_id=<initiator_id>'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
--header 'secret-key-timestamp: <timestamp>'
```

Success response (list truncated)

```
{
  "response_status_id": 0,
  "param_attributes": {
    "list_elements": [
      { "label": "Agra",          "value": "Agra-DVVNL" },
      { "label": "Lucknow",      "value": "Lucknow-MVVNL" },
      { "label": "Kanpur Nagar", "value": "Kanpur Nagar-KESCO" },
      { "label": "Varanasi",     "value": "Varanasi-PUVNL" }
      /* ... additional districts ... */
    ]
  },
  "response_type_id": 2110,
  "message": "Success",
  "status": 0
}
```

Response fields

Field	Description
label	District name to show the agent.
value	district_discome value to pass to Fetch Bill / Pay Bill (operator 190).

Response — districts unavailable

```
{
  "response_status_id": 0,
  "response_type_id": 2434,
  "message": "Failed to fetch districts",
  "status": 0
}
```

4. Recharge Lookups

For prepaid mobile and DTH recharges, use these helpers to detect the operator/circle from a mobile number and to list available plans. The recharge itself is then submitted through Pay Bill (Section 6).

4.1 Get Recharge Operator

GET /customer/payment/bbps/recharge/{mobile}/operator?initiator_id={initiator_id}

Detects the telecom operator and circle for a given mobile number. The returned dependent_params (e.g. phone_operator_code, circle_area) feed into the plans lookup.

Sample request

```
curl --location 'https://api.eko.in:25002/ekoicici/v3/customer/payment/bbps/recharge/{mobile}/operator?initiator_id=<initiator_id>'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
--header 'secret-key-timestamp: <timestamp>'
```

Success response

```
{
  "response_status_id": 0,
  "dependent_params": [
    { "name": "phone_operator_code", "value": 1 },
    { "name": "circle_area", "value": "23" }
  ],
  "data": { "updates": "" },
  "response_type_id": 1804,
  "message": "success",
  "status": 0
}
```

4.2 Get Recharge Plans

GET /customer/payment/bbps/recharge/{mobile}/operator/plans?initiator_id={initiator_id}&phone_operator_code={code}&circleid={circle}

Lists available recharge plans for the operator and circle. Plans are grouped with header metadata describing the fields (amount, validity, description).

Query parameters

Parameter	Type	Required	Description
initiator_id	string	Yes	Partner initiator ID.
phone_operator_code	string	Yes	Operator code from Get Recharge Operator.
circleid	string	Yes	Circle identifier.

Sample request

```
curl --location 'https://api.eko.in:25002/ekoicici/v3/customer/payment/bbps/recharge/{mobile}/operator/plans?initiator_id=<initiator_id>&phone_operator_code=400&circleid=5'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
--header 'secret-key-timestamp: <timestamp>'
```

Success response (plans truncated)

```
{
  "response_status_id": 0,
```

```

"dependent_params": [
  {
    "type_metadata": [
      {
        "headers": [
          { "name": "amount", "value": "price" },
          { "name": "validity", "value": "validity" },
          { "name": "plan_description", "value": "plan_description" }
        ],
        "dependent_params": [ { "name": "amount", "value": "amount" } ]
      }
    ],
    "name": "req_list",
    "value": [
      { "amount": "224", "validity": "30 Days", "plan_description": "Unlimited Calls + 4GB
data ..." },
      { "amount": "349", "validity": "28 Days", "plan_description": "1.5 GB/day + Unlimited
Calls ..." }
      /* ... additional plans ... */
    ]
  }
],
"response_type_id": 1804,
"message": "success",
"status": 0
}

```

Response — plans unavailable

```

{
  "response_status_id": 0,
  "data": { "message": "Unable to fetch plans" },
  "response_type_id": 1805,
  "message": "fail to fetch details",
  "status": 0
}

```

5. Fetch Bill

GET /customer/payment/bbps/bill?
initiator_id={initiator_id}&phone_operator_code={op}&utility_acc_no={acc}&...

Retrieves the live bill amount and customer name for a biller. Call this before Pay Bill so the agent can confirm the amount. Parameters are passed in the query string.

Request parameters

Parameter	Type	Required	Description
initiator_id	string	Yes	Partner initiator ID.
phone_operator_code	string	Yes	operator_id from Get Operators.
utility_acc_no	string	Yes	Bill / account number for the biller.
confirmation_mobile_no	string	Yes	Customer mobile number linked with the bill.
sender_name	string	Yes	Customer name.
category	integer	Yes	operator_category_id from Get Categories.
client_ref_id	string	Yes	Unique partner reference for this request.
source_ip	string	Yes	Originating client IP address.
district_discome	string	No	Required for operator 190 only (Section 3).
state	string	No	State code, where the operator requires it.

Parameters vary by operator

The fields above are the common set. The actual input fields depend on the operator — call Get Operator Parameters (Section 2.4) first and treat its `list_elements` as the source of truth. Any additional parameters it returns (using each `param_name`) must also be passed here in Fetch Bill **and** in Pay Bill (Section 6). The same parameter set must be sent to both calls.

Sample request — without district discome

```
curl --location 'https://api.eko.in/ekoicici/v3/customer/payment/bbps/bill?
initiator_id=<initiator_id>&phone_operator_code=53&confirmation_mobile_no=9903457748&utility_acc_n
o=3287820071&sender_name=Asaad&category=8&client_ref_id=<unique_client_ref_id>&source_ip=<client_i
p>'
--header 'Content-Type: application/json'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
--header 'secret-key-timestamp: <timestamp>'
```

Sample request — with district discome (operator 190)

```
curl --location 'https://api.eko.in/ekoicici/v3/customer/payment/bbps/bill?
initiator_id=<initiator_id>&phone_operator_code=190&confirmation_mobile_no=9510251564&utility_acc_
no=3528689000&sender_name=Ram
%20Prasad&state=1&category=8&client_ref_id=<unique_client_ref_id>&source_ip=<client_ip>&district_d
iscome=Lucknow-MVVNL'
--header 'Content-Type: application/json'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
--header 'secret-key-timestamp: <timestamp>'
```

Success response

```
{
  "response_status_id": -1,
  "data": {
    "amount": "2870.0",
```

```

    "utilitycustomername": "Mohammad Saad Mohd Tarib Saddiqui",
    "ifsc_status": 1,
    "user_code": "35739001",
    "customer_id": "3287820071",
    "billDueDate": "20260728",
    "billdate": "null",
    "billDetailsList": [],
    "bbpstrxnrefid": "",
    "billername": ""
  },
  "response_type_id": 1052,
  "message": "Due Bill Amount For utility",
  "status": 0
}

```

Response fields

Field	Description
amount	Bill amount due — present this and reuse it at Pay Bill.
utilitycustomername	Customer name on the bill — echo as utilitycustomername on Pay Bill.
billDueDate	Due date (YYYYMMDD).
customer_id	Account / consumer number resolved by the biller.
bbpstrxnrefid	BBPS transaction reference, where provided.

Error — missing required field

```

{
  "response_status_id": 1,
  "invalid_params": { "utility_acc_no": "Please provide the value of the field {2} {3}" },
  "response_type_id": -1,
  "message": "Please provide the value of the field",
  "status": 97
}

```

Error — bill fetch failed

```

{
  "response_status_id": 1,
  "data": { "reason": "Please try again after some time." },
  "response_type_id": 1468,
  "message": "Unable to fetch bill",
  "status": 1468
}

```

Field mapping into Pay Bill

Carry amount and utilitycustomername from this response into the Pay Bill request, and reuse the same client_ref_id.

6. Pay Bill

POST /customer/payment/bbps

Pays the bill (or submits a recharge). Send the amount confirmed at Fetch Bill and the utilitycustomername returned there. For operator 190, include district_discome.

Request parameters

Parameter	Type	Required	Description
initiator_id	string	Yes	Partner initiator ID.
phone_operator_code	string	Yes	operator_id from Get Operators.
utility_acc_no	string	Yes	Bill / account number.
confirmation_mobile_no	string	Yes	Customer mobile number.
sender_name	string	Yes	Customer name.
category	integer	Yes	operator_category_id.
amount	number	Yes	Amount to pay (from Fetch Bill).
utilitycustomername	string	Yes	Customer name from Fetch Bill.
client_ref_id	string	Yes	Unique partner reference (reuse from Fetch Bill).
source_ip	string	Yes	Originating client IP address.
district_discome	string	No	Required for operator 190 only.

Parameters vary by operator

As with Fetch Bill, the fields above are the common set. Pass every parameter returned by Get Operator Parameters (Section 2.4) for the chosen operator, using each param_name. The parameter set sent to Pay Bill must match the set sent to Fetch Bill for the same operator.

Sample request — without district discome

```
curl --location 'https://api.eko.in/ekoicici/v3/customer/payment/bbps'
--header 'Content-Type: application/x-www-form-urlencoded'
--header 'developer_key: <developer_key>'
--header 'secret-key: <secret-key>'
--header 'secret-key-timestamp: <timestamp>'
--data-urlencode 'initiator_id=<initiator_id>'
--data-urlencode 'phone_operator_code=53'
--data-urlencode 'confirmation_mobile_no=9903457748'
--data-urlencode 'utility_acc_no=3287820071'
--data-urlencode 'sender_name=Asaad'
--data-urlencode 'category=8'
--data-urlencode 'client_ref_id=<unique_client_ref_id>'
--data-urlencode 'source_ip=<client_ip>'
--data-urlencode 'utilitycustomername=Mohammad Saad Mohd Tarib Saddiqui'
--data-urlencode 'amount=2870'
```

Sample request — with district discome (operator 190)

```
# Same as above, plus:
--data-urlencode 'phone_operator_code=190'
--data-urlencode 'district_discome=Lucknow-MVVNL'
```

Success response

```
{
  "response_status_id": 0,
  "data": {
```

```

    "tx_status": "0",
    "txstatus_desc": "Success",
    "status_text": "SUCCESS",
    "tid": "3570553488",
    "amount": "2870.0",
    "totalamount": "2870.0",
    "fee": "0.0",
    "tds": "0.024",
    "commission": "1.2",
    "balance": "46537.3",
    "user_code": "35739001",
    "operator_name": "B.E.S.T Mumbai",
    "utilitycustomername": "Mohammad Saad Mohd Tarib Saddiqui",
    "customermobilenum": "9903457748",
    "payment_mode_desc": "CASH",
    "last_used_okekey": "203",
    "account": "3287820071",
    "timestamp": "2026-07-25T09:52:57.428Z"
  },
  "response_type_id": 333,
  "message": "Success Last_used_0keyKey: 203",
  "status": 0
}

```

Key response fields

Field	Description
tx_status	Transaction status (0 = success).
tid	Eko transaction ID — store for reconciliation.
amount / totalamount	Amount paid.
fee / tds / commission	Fee, TDS, and agent commission.
balance	Agent wallet balance after payment.
operator_name	Biller the payment was made to.
approvalreferencenumber	Biller approval reference, where provided.

Error — amount mismatch

```

{
  "response_status_id": 1,
  "data": { "last_used_okekey": "", "tid": "3571183956" },
  "response_type_id": 208,
  "message": "utility.payment.failed Amount entered does not match with bill amount. Please try again",
  "status": 208
}

```

Error — missing required field

```

{
  "response_status_id": 1,
  "invalid_params": { "amount": "Please provide the value of the field {2} {3}" },
  "response_type_id": -1,
  "message": "Please provide the value of the field",
  "status": 97
}

```

Amount must match the fetched bill

Pay the exact `amount` returned by Fetch Bill. A mismatch is rejected with status 208. Always persist `tid` and your `client_ref_id` for reconciliation before retrying.

7. Response Status Reference

All responses share a common envelope. Note that `response_status_id` may be 0, -1, or 1 across these APIs (lookups and bill fetch often return -1 on success); rely on `status` and the documented `response_type_id` to branch. Handle the following explicitly.

status	Meaning	Action
0	Success (see <code>response_type_id</code> for the specific outcome).	Proceed per the flow.
97	A required field is missing.	Supply the named field and retry.
208	Payment failed — amount does not match the bill.	Re-fetch the bill and pay the exact amount.
319	Invalid sender / initiator.	Check <code>initiator_id</code> is registered.
1468	Unable to fetch bill.	Retry after some time; verify account details.

Key `response_type_id` values

2457 categories · 2459 locations · 2461 operators · 2463 operator parameters · 1804 recharge operator/plans · 1805 plans unavailable · 1052 bill fetched · 333 payment successful · 208 amount mismatch · 1468 bill fetch failed.

Reconciliation

Always persist `tid` and your `client_ref_id` for every payment to support settlement and dispute resolution.

Appendix A — Category Reference

Common BBPS categories and their identifiers (from Get Categories). The live list is authoritative; use it at runtime rather than hard-coding.

ID	Category	ID	Category
1	Broadband Postpaid	14	Education
2	Gas	15	Municipal Taxes
4	DTH	16	Clubs and Associations
5	Mobile Prepaid	17	Cable TV
6	Tax	18	LPG Cylinder
7	Credit Card	19	Hospital
8	Electricity	20	Insurance
9	Landline Postpaid	21	Loan
10	Mobile Postpaid	22	FASTag
11	Water	23	Municipal Services
12	Housing Society	24	Rental Payment
13	Subscription	27+	eChallan, Agent Collection, EV Recharge, ...

Appendix B — Reference Links

Resource	Link
Developer portal	https://developers.eko.in
BBPS reference	https://developers.eko.in/reference

All keys, credentials, and personal identifiers shown in this document are sample or placeholder data for illustration only. Contact your Eko integration representative for production credentials and support.